1. Signed trace projectors for a prime-index Pell squarefull gate

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Summary. Let $(1+\sqrt{2})^n = A_n + B_n\sqrt{2}$. We give an exact signed-trace encoding of the still open assertion that $A_\ell B_\ell$ is not squarefull at an odd prime index. At every odd index,

$$A_{2n} - 1 = 2A_n^2, \qquad A_{2n} + 1 = 4B_n^2.$$

Thus squared support in either coordinate is equivalent to fourth-power support in the corresponding adjacent trace factor. We also identify a canonical Chinese-remainder projector $E_n = (A_{2n}+1)/2 = 2B_n^2$ whose exact idempotent defect is $2(A_nB_n)^2$, and show that one explicit Newton correction raises its precision from $(A_nB_n)^2$ to $(A_nB_n)^4$.

An exhaustive computation over odd prime $\ell \le 800000$ and support prime $q \le 2000000$, independently replayed by matrix powering, finds exactly two depth-two support collisions, one in each channel. Both give signed-trace depth exactly four and therefore disprove several tempting strengthenings. The index-seven row is nonsquarefull because its opposite coordinate is the exponent-one prime 239; the global squarefull status of the large row is unresolved. Neither row is a counterexample to the *abc* conjecture. All algebraic claims and the small counterexample are kernel checked in Lean.

1.1. The Pell coordinates and the unresolved gate.

Define nonnegative integers A_n, B_n by

$$(1+\sqrt{2})^n = A_n + B_n\sqrt{2}.$$

Equivalently, $(A_0, B_0) = (1, 0)$ and

$$(A_{n+1}, B_{n+1}) = (A_n + 2B_n, A_n + B_n).$$

The norm identity is

$$A_n^2 - 2B_n^2 = (-1)^n. \tag{1}$$

For an integer $N > 0$, *squarefull* means that $p \mid N$ implies $p^2 \mid N$ for every prime p.

The present work begins with the following fixed-parameter result from the companion transversality analysis.

Proposition 1.1 (fixed-parameter gate). *Let ℓ be an odd prime. Every support prime of A_ℓ and B_ℓ is transverse at the Pell polynomial parameter $T = 2$. Furthermore, all first Hensel displacements vanish if and only if $A_\ell B_\ell$ is squarefull.*

Proposition 1.1 is recalled as an input, rather than silently being folded into a computation. Classical rank-of-apparition theory for Lucas sequences supplies the necessary candidate classes $q \equiv \pm 1 \pmod{2\ell}$ used below; primitive-divisor results such as [1] provide the broader historical setting. The exclusion of a squarefull product at every odd prime index remains open.

1.2. Depth doubling in adjacent signed trace factors.

Lemma 1.2 (addition law). *For all $m, n \ge 0$,*

$$A_{m+n} = A_mA_n + 2B_mB_n,$$
$$B_{m+n} = A_mB_n + B_mA_n.$$

Proof. Multiply $(A_m + B_m\sqrt{2})(A_n + B_n\sqrt{2})$ and compare the rational and $\sqrt{2}$ coefficients. This is also an induction using the defining recurrence. □

Put $Z_n = A_{2n}$. Taking $m = n$ in Lemma 1.2 gives

$$Z_n = A_n^2 + 2B_n^2, \qquad B_{2n} = 2A_nB_n.$$

Theorem 1.3 (signed trace-square identities). *If n is odd, then*

$$Z_n - 1 = 2A_n^2, \qquad Z_n + 1 = 4B_n^2.$$

Proof. For odd n, equation (1) gives $2B_n^2 = A_n^2 + 1$. Substitute this identity into $Z_n = A_n^2 + 2B_n^2$. □

Theorem 1.4 (exact depth dictionary)**.** *Let n be odd.*

(1) *If q is an odd prime dividing A_n, then*
$$q^4 \mid Z_n - 1 \iff q^2 \mid A_n.$$
(2) *If q is an odd prime dividing B_n, then*
$$q^4 \mid Z_n + 1 \iff q^2 \mid B_n.$$

More precisely,
$$v_q(Z_n - 1) = 2v_q(A_n), \qquad v_q(Z_n + 1) = 2v_q(B_n)$$
in the applicable channel. Hence coordinate depth exactly two gives signed-trace depth exactly four.

Proof. Use Theorem 1.3. The factors 2 and 4 are units at an odd prime, so taking q-adic valuations gives the asserted equalities and all their stated consequences. ☐

For odd integers A, B and a trace value Z satisfying the two identities of Theorem 1.3, define
$$\mathrm{ST}_4(A, B, Z) \iff \begin{cases} q^4 \mid Z - 1 & \text{for every prime } q \mid A, \\ q^4 \mid Z + 1 & \text{for every prime } q \mid B. \end{cases}$$

Corollary 1.5 (exact signed-packet gate)**.** *For every odd prime ℓ,*
$$\begin{aligned} &\textit{all fixed-}T = 2 \textit{ first Hensel displacements vanish} \\ \iff\ & A_\ell B_\ell \textit{ is squarefull} \\ \iff\ & \mathrm{ST}_4(A_\ell, B_\ell, A_{2\ell}). \end{aligned}$$

Proof. The first equivalence is Proposition 1.1. The coordinates are coprime, so their product is squarefull exactly when both coordinates are squarefull. Apply Theorem 1.4 to every support prime. ☐

Corollary 1.6 (radical compression)**.** *If both coordinates at an odd index are squarefull, then*
$$\mathrm{rad}(A_n)^4 \mid Z_n - 1, \qquad \mathrm{rad}(B_n)^4 \mid Z_n + 1,$$
and
$$\mathrm{rad}(A_n B_n)^4 \mid (Z_n - 1)(Z_n + 1) = 8(A_n B_n)^2.$$

Proof. Squarefullness gives $\mathrm{rad}(A_n)^2 \mid A_n$ and $\mathrm{rad}(B_n)^2 \mid B_n$. Square these divisibilities and use Theorem 1.3. Coprimality combines the radicals, while direct multiplication gives the displayed product identity. ☐

The last divisibility is compatible with squarefullness and is not a contradiction. Its content is the placement of the two complete support packets in adjacent factors of one trace coordinate.

1.3. **The exact projector and its Newton continuation.** Fix an odd index and abbreviate $A = A_n$, $B = B_n$, $U = AB$, and $Z = Z_n$. Define
$$E = \frac{Z + 1}{2} = 2B^2 = A^2 + 1.$$

Theorem 1.7 (channel projector)**.** *The integer E satisfies*
$$E \equiv 1 \pmod{A^2}, \qquad E \equiv 0 \pmod{B^2},$$
and has exact defect
$$E^2 - E = 2U^2. \tag{2}$$
Thus E is idempotent modulo U^2. If $U \geq 3$, it is not idempotent modulo U^3.

Proof. The congruences follow from the two expressions for E. Their product gives
$$E(E - 1) = 2B^2 A^2 = 2U^2.$$
If U^3 divided the last expression, then U would divide 2, which is impossible for $U \geq 3$. ☐

The sharpness of the raw projector does not terminate the construction. For integers e, d, set

$$\mathcal{N}(e,d) = e - (2e-1)d.$$

Theorem 1.8 (one Newton step). *If $e^2 - e = d$, then*

$$\mathcal{N}(e,d)^2 - \mathcal{N}(e,d) = d^2(4d-3).$$

For $d = 2U^2$, the corrected projector is congruent to e modulo U^2 and is idempotent modulo U^4.

Proof. Write $z = 2e - 1$. Since $z^2 = 4d + 1$, expansion yields

$$\begin{aligned}(e-zd)^2 - (e-zd) &= d - z^2 d + z^2 d^2 \\ &= d^2(4d-3).\end{aligned}$$

The correction zd is divisible by U^2, and the new defect contains $d^2 = 4U^4$. □

1.4. **Complete-premise finite counterexamples.** The search enumerates every odd prime $\ell \le 800000$ and every prime $q \le 2000000$ in the necessary classes $q \equiv \pm 1 \pmod{2\ell}$. It computes (A_ℓ, B_ℓ) modulo q^2, and recomputes every repeated hit modulo q^5. The producer uses binary powering in $\mathbb{Z}[\sqrt{2}]$. A verifier independently powers

$$\begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}^{\ell}.$$

A separate certificate checks primality by complete trial division and recomputes each collision by a third, linear-recurrence algorithm.

Proposition 1.9 (replayed finite search). *In the stated rectangle there are* 63950 *prime indices,* 764366 *candidate-prime tests, and* 12356 *actual support hits. Exactly two support hits have depth at least two. They are the rows in Table 1; both have coordinate depth exactly two and signed trace depth exactly four. No depth-three hit and no index with repeated support in both channels occurs in the rectangle.*

Table 1. The complete repeated-support rows. The two quotient columns are nonzero residues modulo q; C_ℓ denotes the indicated channel and $S_\ell = A_{2\ell} \mp 1$ its signed trace shift.

ℓ	q	ch.	C_ℓ/q^2	$C'_\ell(2)$	S_ℓ/q^4
7	13	B	1	12	4
773231	1546463	A	1090979	326969	1407445

Proof. The exhaustive producer and independent matrix replay agree on every count and on both complete rows. For the larger row, complete trial division through 879 and 1243 proves the primality of ℓ and q, respectively. Three orbit algorithms agree modulo $q^5 = 8844996565598309452666138088543$. The nonzero residues in Table 1 prove $q^2 \parallel C_\ell$, transversality of the fixed parameter, and $q^4 \parallel A_{2\ell} \mp 1$. □

The first row can also be read without software:

$$B_7 = 169 = 13^2, \qquad A_{14} + 1 = 114244 = 4 \cdot 13^4.$$

It follows that $13^5 \nmid A_{14} + 1$.

Corollary 1.10 (refuted strengthenings). *Each of the following universal statements is false:*

1. *no support prime repeats at an odd prime index;*
2. *repetition is confined to the A channel;*
3. *repetition is confined to the B channel;*
4. *the minimal representative $q = 2\ell \pm 1$ forces simple support;*
5. *coordinate depth two always promotes to signed-trace depth five.*

Proof. The two rows of Table 1 meet every premise of the relevant claims. The first supplies the B-channel counterexample, the second the A-channel counterexample, and both use the minimal residue representative. Exact trace depth four refutes the last assertion. □

Remark 1.11 (logical boundary). The two rows are local collisions. Neither row says that both coordinates at its index are squarefull. Consequently neither refutes the signed-packet exclusion in Corollary 1.5, and neither is an *abc* counterexample. Bounded absence of a simultaneous row is recorded only as a finite observation.

1.5. **Formal and computational record.** The Lean module `PellSignedTraceProjector 20260903.lean` formalizes the addition and doubling laws, Theorems 1.3–1.8, the radical consequences, and the complete index-seven refutation. It compiles with `-DwarningAsError=true`. Its companion audit prints the axioms of every declaration. The union consists only of the standard imported library axioms

`propext`, `Classical.choice`, `Quot.sound`.

The reproducibility bundle contains the producer, matrix verifier, complete-premise collision certifier, JSON outputs, exit codes, compiler logs, environment metadata, and SHA-256 manifests. No finite no-hit is promoted to a global theorem.

1.6. **Remaining problem.** The prime-index squarefull gate is now the existence problem for two fourth-power support packets in the adjacent factors $A_{2\ell}-1$ and $A_{2\ell}+1$, together with their exact product and a canonical tower of corrected projectors. The depth-two collisions prove that neither a single channel nor one additional trace digit can close the problem. A successful continuation must couple all support primes across both adjacent factors, or construct a genuine simultaneous packet. Since no complete-premise counterexample to that global exclusion is presently known, the route remains open.

References

[1] Y. Bilu, G. Hanrot, and P. M. Voutier, *Existence of primitive divisors of Lucas and Lehmer numbers*, J. Reine Angew. Math. **539** (2001), 75–122.